

Assumptions

1. An AUTHORITY manages at least one STREET.
2. An OWNER owns at least one PROPERTY and a PROPERTY can be owned by at least one OWNER.
3. An AUTHORITY signs at least one CONTRACT.
4. Every WASTE_TYPE may have zero or many CONTRACT_WASTES and BIN_TYPES.
5. A CONTRACT covers at least one CONTRACT_WASTE.
6. A CONTRACT_WASTE may have zero or many CONTRACT_BINS.
7. A physical bin can be supplied if its specific waste type and size was explicitly approved under a specific contract.
8. A CONTRACT_WASTE may have zero or many collection runs (COLL_RUN).
9. A TRUCK works for zero or many collection runs (COLL_RUN).
10. A BIN may have zero or many COLLECTIONS.
11. A collection run (COLL_RUN) may collect zero or many times of COLLECTION.
12. A surrogate key prop_id in PROPERTY entity was introduced as the primary key in the logical model to avoid excessively large and complex foreign keys propagating into child entities as well as improve database efficiency. A unique constraint has been applied to the original natural keys combination, which is auth_name, street_id, and prop_street_num.
13. A surrogate key collrun_id in COLL_RUN entity was introduced as the primary key in the logical model to avoid excessively large and complex foreign keys propagating into child entities as well as improve database efficiency. A unique constraint has been applied to the original natural keys combination, which is collrun_date, contract_num, and waste_type_id.
14. A surrogate key cb_num in CONTRACT_BIN entity was introduced as the primary key in the logical model to avoid excessively large and complex foreign keys propagating into child entities as well as improve database efficiency. A unique constraint has been applied to the original natural keys combination, which is contract_num, waste_type_id, and bin_size.
15. In the logical model, the attributes of the BIN_STATUS (bs_collected_kgs and bs_overweight) has been merged into the COLLECTION. It is assumed that these attributes are fully functionally dependent on the specific collection event, separating them into 1:1 relationship would cause unnecessary database fragmentation.
16. A surrogate key cw_num in CONTRACT_WASTE entity was introduced as the primary key in the logical model to resolve the issue of converging foreign keys and preventing a duplicate waste_type_id attribute from propagating down into CONTRACT_BIN. Therefore, CONTRACT_BIN no longer inherits waste_type_id directly from the contract relationship. The database schema alone can no longer natively enforce the rule that a supplied bin strictly matches the contracted waste type. So it is assumed that the application layer will enforce this business rule by cross-checking the waste type associated with the CONTRACT_WASTE record against the BIN_TYPE record before permitting any insert operation into CONTRACT_BIN table.